

Reductionism and Earth Science

the interdisciplinary study of natural phenomena and the origin of knowledge

in 2270 words

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1. Introduction

In this essay, a viewpoint on reductionism and Earth Science will be proposed. The central question in this essay is: "Can Earth science be reduced to other scientific disciplines such as physics, chemistry or biology?" To provide a framework, the main influences on the subject will be presented by discussing some of the historical philosophical views that relate to them in the chapter 'philosophical views'.

After that, several topics will be re-examined to elucidate a central idea in the chapter 'analysis'. First, the nature of science in simplified form as the interaction of the mind with reality will be explored. This will lead to a re-appraisal of scientific disciplines. Reductionism in science will then be discussed and its definition refined. From there, interdisciplinary work will be described and its value in relation to scientific disciplines and reductionism. The idea of interdisciplinary work will then be discussed in application to natural phenomena. From this, the essay proposes a new understanding of scientific progress and reductionism in which the interdisciplinary study of natural phenomena is the research that is closest to the origin of knowledge.

Finally, the conclusion will present the answer to the question posed in this introduction using the results of the analysis.

2. Philosophical views

2.1 Scientific realism

The essay builds on the concept of realism similar to that of scientific realism as put forward by Godfrey-Smith (2003). This type of realism is defined as an assumption that reality exists, common to all people, and largely independent of our thoughts and actions, except for certain parts that are influenced by us, such as language and architecture, etc. It furthermore states that the aim of science is to provide accurate representations of what reality is like, including the parts of reality that are unobservable, e.g. have not been observed yet. To represent reality in a way that we can understand it, science uses different simplified concepts that refer to a part of reality (linguistic, mathematical, symbolic concepts) and build with these a structure that has a similarity to the part of reality that it is meant to explain.

2.2 Explanation

There are several theories about how science explains reality. Whether this is by deriving an event or phenomenon from initial conditions and one or more natural laws (covering law of explanation), or by showing how it is caused by known entities (theory of causation), or how certain principles account for the nature of multiple issues including the one explained (unification theory), or a pluriform theory wherein many different methods that determine what an adequate explanation is exist across scientific disciplines is not important for this essay. A concept used in the discussion of explanation is, that of explanatory inference. It is the way in which a hidden pattern or law is extracted from a collection of information and is supposed to be the principle by which the information can be explained. In a way, this method tries to reduce the amount of information while retaining most of the knowledge within it.

2.3 Reductionism

The principle of reductionism has been defined by Nagel (1961) as: "the explanation of a theory or set of experimental laws established in one area of inquiry, by a theory usually though not invariably formulated for some other domain".

Reductionism then is the process of reducing theories to more essential laws that can still explain the functions of the original theory. This process can re-occur several times, and thoughts have been expressed about whether an ultimate theory of truth can in this way be found.

3. Analysis

Scientific realism is based on many assumptions about how we experience reality. To provide a logical progression to the main idea in this essay, a quick redefinition of this process will be presented. It is largely identical to scientific realism but it focuses on certain aspects that will be expanded on in the next chapters.

3.1 Scientific realism, explanation and reduction

As we perceive the world around us, we make mental representations of things. Both elements of experience such as physical objects and events, as our response to them, perceived regularities in their occurrence, possibilities of manipulation, reproductions and more become conceptions in our mind. Because we can not observe, or process, the whole of reality in our minds, the concepts we form are representations. They are meant to express a relevant characteristic of what they refer to, allowing us to concentrate on those parts of reality that are of importance to us.

The nature of these conceptions is determined by the interaction with the world around us and the inner characteristics of our mind. The former entails both the domain and sensitivity of our senses as well as the necessity of our interaction (it is not unreasonable to assume the first mental developments focused on food and safety), the latter our ability to comprehend, process and reproduce these experiences in our mind. From the conceptions we form, new action results and this redefines our interaction with our world. Mental development, the forming of conceptions, and our interaction with the world develop in relation to each other.

In science the same process occurs. We have expanded the scope of our interaction with the world and sensory information with technology and human development (telescopes and athletics for instance), as well as our interpretation abilities (by arts, technical training, etc). Besides our personal exploration of the world and the development of mental concepts of it, we have collective efforts that attempt to reach a deep understanding of our world and higher mental development. These efforts or disciplines have well-established methods of observation and experiment, typical frameworks of interpretation, and habitual languages, symbolisms in which they express their findings, although these develop over time and variations may occur. These disciplines are sciences such as physics, biology or chemistry.

They form conceptions, in this case fundamental concepts, laws, theories, that are determined by the method of interaction with the world and the manner in which these conceptions are processed, and that determine further research.

The way in which we form mental concepts to represent relevant aspect of reality also has its counterpart in science. While for a person, subjective criteria might be a priority, science often seeks to explain and understand reality. Scientific disciplines attempt to find ways to represent reality in terms that represent the actual workings of reality itself, that is to discover governing principles in nature. Whether they are patterns of mechanical functionality, or relative influences in a dynamic system, or principles that govern the emergence of patterns at different scales in a holistic view, and whether they represent these or refer to actual essences responsible for them is not important for this essay, only that they can account for what has been observed in reality.

To propose a set of essential laws that can explain a larger theory, or reductionism, is very similar to the first principle of experience or science: to represent a complex amount of information by its 'essential' pattern or functionality that governs it.

Each discipline has a different aim, different focus, and a different method of research. They have different and often unique essential concepts to account for the parts of reality they study.

3.2 Scientific disciplines and interdisciplinary science

Since disciplines remain mental constructs for us to help understand and study our world, they are a bridge between our mind and the nature we study. By comparing different disciplines, we can generate some understanding of what a discipline is and how we observe and study nature. If we understand what observation and conceptualization is, the need for a disciplinary framework can ultimately be discarded. The reconstruction of observed or inferred patterns and governing principles in language and mathematics is projecting information in nature into lesser forms. The thoughts behind these representations can be as similar to nature as we can conceive of them, the way in which we express them is not. As it is only the origin of this representation, or the actual phenomenon, which matters, outlining it not as a definite form but by stating the parameters of its form might be more helpful. Some truths can not be condensed to fit the framework of our limited language and mathematics – they mind itself struggles to conceive these ideas. By using concepts and principles from all manner of disciplines, one can give directions to the mind how to conceive of such a principle. This is interdisciplinary science. It allows for the expression of

governing elements, theories, progressions, that our scientific language is not yet capable of expressing. It is also one step closer to the creative process of the mind, that generates thoughts and ideas before condensing them into language. It removes the constraining framework of a specific discipline and can discover the need for different disciplines or subbranches of existing disciplines in order to express previously unexamined characteristics of elements of study. When a concept can be adequately described by a particular discipline, effort can be undertaken to describe it with another discipline and compare the two, and once again open the research to exploration of the observations or knowledge in their pre-disciplinary form, and the as yet undefined and unknown areas of any concept so balanced between disciplinary frameworks.

The eventual result of this approach is to provide disciplines with a wider understanding of, more topics in and new methods of research into their subjects, and possibly to generate new disciplines. An expression in language or mathematics is desired in order to perform structural logical or calculative operations on what has been discovered. But although the application of a rigid discipline may greatly enhance the level of detail on any knowledge, condensing the original concept into language represents an inevitable loss of information.

3.3 Earth science

To apply an interdisciplinary method to any concept or area of study can provide a great deal of information about the nature of the subject at hand, and how our minds conceive and process information both in the disciplines as during the interdisciplinary work. If applied to reality, it can also further progress into the understanding of nature. As we defined scientific realism as a view wherein both we ourselves and a common reality exists, these two subjects are the most direct sources of knowledge. Research of subjects derived from these two represent a selective choice and thereby the influence of our mind. Charting how these and other choices influence the information we find, how the interaction of our mind with reality influences both, and how it influences the development of the discipline is a difficult task. It tells us much about our place in the world but it is not the best way to understand the world or our minds themselves.

Of these two subjects, the study of the world has been the focus of this essay. The most 'pristine' reality is the part of our world that we experience but have not yet influenced, the nature of Earth. This is true both for the study of the nature of reality, as the nature of the effects of our interaction with it since we can observe them in their first stages.

The interdisciplinary approach, combined with disciplines to further study particular elements to be returned to the interdisciplinary process, is the method that allows for closest examination of information in research.

Therefore, an interdisciplinary study of nature is the best source for and the best method to obtain knowledge about independent reality, and partly about the influence we have on reality. This study is called Earth Science.

4. Conclusion

Reductionism is defined as reducing a theory to a more essential set of laws. This means to find the essence that governs, or determines, the reduced information. The question to be answered in this essay was whether Earth science can be reduced to other scientific disciplines such as physics, chemistry and biology. In other words: if the information in Earth science can be reduced to essential principles in these disciplines.

Because Earth science is an interdisciplinary science, studying concepts between the different disciplines, they can not be reduced to concepts within any of these disciplines without losing information.

Furthermore, since reductionism is the reducing of theories to their essential components, and any theory is a mental conception of a certain phenomenon, we could argue that the essence of any theory in any discipline is the pattern that it is meant to represent. These have a higher information value, they are the origin of the theories and can explain them, and they have a structure that is more concise and condensed than any language that attempts to describe it.

Insofar as physics, chemistry and biology study nature, and considering Earth science is the study of the patterns that are the essences of the theories in these disciplines, it seems more reasonable that these

disciplines can be reduced to principles in Earth science, that is, interdisciplinary concepts of natural phenomena.

However, these disciplines also consist of the influence of the mind in the form of a selective approach to what natural phenomena to study, particular methodologies for obtaining and processing information and languages of expression. This influence and its effect on the theories of the disciplines, as well as the study of artificial phenomena, can not be reduced to Earth science alone. They can however be studied with its same methodology, while studying the nature of the interdisciplinary method or of uninfluenced reality can is not possible with any of the discussed disciplines.

Therefore, not only can Earth science not be reduced to any other discipline, it can be used to reduce other disciplines with.

5. Bibliografie

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